

## PUNE, India

WMO: 430630

Lat: 18.533N

Lon: 73.850E

Elev: 558

StdP: 94.80

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |            |            |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|------------|------------|-------------|-----------------------|-------------|------------|-------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |             | 99%        |            |             | 0.4%                  |             | 1%         |             |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP         | HR         | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>10.0 | (c)<br>11.2 | (d)<br>1.5                    | (e)<br>4.5 | (f)<br>31.4 | (g)<br>3.4 | (h)<br>5.1 | (i)<br>30.1 | (j)<br>3.4            | (k)<br>26.1 | (l)<br>3.1 | (m)<br>25.9 | (n)<br>0.0            | (o)<br>90 | (p)<br>0.260 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD<br>to 0.4% DB |            |
|---------------|---------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------------|------------|
|               |                           | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                         |            |
|               |                           | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                    | PCWD       |
| (a)<br>5      | (b)<br>12.5               | (c)<br>38.5     | (d)<br>19.9 | (e)<br>37.4 | (f)<br>19.8 | (g)<br>36.4 | (h)<br>19.9 | (i)<br>24.7         | (j)<br>30.0 | (k)<br>24.3 | (l)<br>29.3 | (m)<br>24.0 | (n)<br>28.7 | (o)<br>1.5              | (p)<br>270 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.5                            | 19.7 | 26.2 | 23.2 | 19.2 | 25.8 | 22.8 | 18.8 | 25.6 | 78.8          | 30.2 | 77.0 | 29.6 | 75.5 | 28.8 | 29.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |      |  |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|------|--|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |      |  |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |      |  |
| (a) 5.0           | (b) 3.9 | (c) 3.3 | (e) 7.6                    | (f) 40.5 | (g) 1.1            | (h) 0.8 | (i) 6.8                                            | (j) 41.1 | (k) 6.2    | (l) 41.6 | (m) 5.5    | (n) 42.1 | (o) 4.7    | (p) 42.7 |      |  |
|                   |         |         | DB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |      |  |
|                   |         |         | WB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |      |  |
|                   |         |         |                            | 6.6      | 26.0               | 1.1     | 1.1                                                | 5.8      | 26.8       | 5.1      | 27.5       | 4.5      | 28.1       | 3.7      | 29.0 |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 25.0   | 20.6  | 22.8  | 26.1  | 29.2  | 29.9  | 27.3  | 25.0  | 24.5  | 25.1  | 25.2  | 22.9  | 20.9 |
|                                                                                  | DBStd       | 3.27   | 1.82  | 1.94  | 2.05  | 1.71  | 1.44  | 1.95  | 1.18  | 0.93  | 1.12  | 1.50  | 1.86  | 1.89 |
|                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD18.3     | 6      | 4     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2    |
|                                                                                  | CDD10.0     | 5463   | 327   | 359   | 500   | 577   | 618   | 518   | 466   | 451   | 453   | 471   | 386   | 337  |
|                                                                                  | CDD18.3     | 2428   | 73    | 127   | 242   | 327   | 359   | 268   | 208   | 193   | 203   | 212   | 136   | 81   |
|                                                                                  | CDH23.3     | 24339  | 909   | 1594  | 3238  | 4504  | 4566  | 2429  | 1083  | 818   | 1188  | 1786  | 1307  | 917  |
|                                                                                  | CDH26.7     | 11007  | 256   | 743   | 1891  | 2747  | 2552  | 891   | 163   | 90    | 292   | 698   | 447   | 237  |
| Wind                                                                             | WSAvg       | 1.0    | 0.4   | 0.6   | 0.7   | 1.0   | 1.8   | 1.9   | 1.7   | 1.5   | 1.0   | 0.5   | 0.4   | 0.3  |
| Precipitation                                                                    | PrecAvg     | 730    | 1     | 1     | 4     | 10    | 32    | 149   | 184   | 136   | 135   | 76    | 25    | 6    |
|                                                                                  | PrecMax     | 1343   | 18    | 31    | 59    | 63    | 149   | 501   | 411   | 378   | 431   | 269   | 177   | 72   |
|                                                                                  | PrecMin     | 384    | 0     | 0     | 0     | 0     | 0     | 12    | 32    | 12    | 23    | 0     | 0     | 0    |
|                                                                                  | PrecStd     | 214    | 3     | 4     | 9     | 15    | 35    | 99    | 92    | 82    | 90    | 69    | 39    | 16   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 31.7  | 35.0  | 38.1  | 39.9  | 39.8  | 36.6  | 31.0  | 30.0  | 32.0  | 33.4  | 32.0  | 31.1 |
|                                                                                  |             | MCWB   | 18.1  | 17.8  | 18.5  | 19.3  | 20.7  | 23.3  | 23.5  | 23.7  | 23.0  | 21.1  | 20.1  | 19.1 |
|                                                                                  | 2%          | DB     | 30.4  | 33.6  | 36.9  | 38.7  | 38.4  | 34.5  | 29.4  | 28.6  | 30.6  | 32.2  | 31.0  | 29.9 |
|                                                                                  |             | MCWB   | 17.9  | 17.7  | 18.5  | 19.4  | 20.8  | 23.4  | 23.4  | 23.3  | 23.5  | 21.1  | 19.9  | 18.7 |
|                                                                                  | 5%          | DB     | 29.3  | 32.4  | 35.8  | 37.7  | 37.2  | 32.9  | 28.4  | 27.7  | 29.5  | 31.3  | 30.2  | 29.0 |
|                                                                                  |             | MCWB   | 17.6  | 17.6  | 18.3  | 19.4  | 21.0  | 23.5  | 23.2  | 23.0  | 23.2  | 21.1  | 19.8  | 18.5 |
|                                                                                  | 10%         | DB     | 28.2  | 31.1  | 34.7  | 36.7  | 35.9  | 31.3  | 27.5  | 26.8  | 28.4  | 30.4  | 29.3  | 28.1 |
|                                                                                  |             | MCWB   | 17.6  | 17.7  | 18.0  | 19.4  | 21.5  | 23.5  | 23.0  | 22.8  | 22.9  | 21.2  | 19.6  | 18.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 20.7  | 20.8  | 22.7  | 23.7  | 25.1  | 25.3  | 24.5  | 24.6  | 25.1  | 24.7  | 23.6  | 22.2 |
|                                                                                  |             | MCDB   | 27.7  | 29.7  | 29.1  | 31.5  | 32.7  | 31.0  | 28.0  | 28.1  | 29.5  | 29.0  | 28.1  | 27.1 |
|                                                                                  | 2%          | WB     | 19.6  | 19.7  | 21.3  | 22.8  | 24.4  | 24.7  | 24.0  | 23.8  | 24.4  | 24.1  | 22.9  | 21.1 |
|                                                                                  |             | MCDB   | 27.4  | 29.5  | 29.6  | 31.1  | 32.0  | 30.3  | 27.3  | 26.7  | 28.5  | 28.4  | 27.6  | 26.7 |
|                                                                                  | 5%          | WB     | 18.8  | 19.1  | 20.4  | 22.1  | 23.9  | 24.3  | 23.6  | 23.5  | 23.9  | 23.7  | 22.3  | 20.3 |
|                                                                                  |             | MCDB   | 27.0  | 29.1  | 30.0  | 31.0  | 31.4  | 29.5  | 26.8  | 26.3  | 27.6  | 27.7  | 26.6  | 26.1 |
|                                                                                  | 10%         | WB     | 18.1  | 18.4  | 19.6  | 21.5  | 23.4  | 23.9  | 23.3  | 23.1  | 23.5  | 23.2  | 21.6  | 19.4 |
|                                                                                  |             | MCDB   | 26.3  | 28.4  | 30.3  | 30.7  | 30.8  | 28.8  | 26.2  | 25.9  | 26.8  | 27.0  | 25.9  | 25.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 16.1  | 17.1  | 17.5  | 15.8  | 12.5  | 7.3   | 4.5   | 4.7   | 7.0   | 11.0  | 13.8  | 15.4 |
|                                                                                  |             | MCDBR  | 17.0  | 18.4  | 18.3  | 16.8  | 14.7  | 10.5  | 6.3   | 6.3   | 9.1   | 12.8  | 14.6  | 16.0 |
|                                                                                  |             | MCWBR  | 6.0   | 5.5   | 5.0   | 4.9   | 4.2   | 2.4   | 2.1   | 2.3   | 3.0   | 3.8   | 5.0   | 5.9  |
|                                                                                  | 5% WB       | MCDBR  | 15.0  | 16.0  | 16.2  | 15.0  | 12.0  | 8.1   | 4.9   | 5.1   | 7.4   | 9.2   | 10.3  | 13.1 |
|                                                                                  |             | MCWBR  | 6.0   | 5.6   | 5.1   | 4.9   | 3.5   | 2.7   | 2.1   | 2.3   | 2.9   | 3.3   | 4.1   | 5.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.424  | 0.431 | 0.499 | 0.550 | 0.543 | 0.578 | 0.576 | 0.544 | 0.522 | 0.554 | 0.494 | 0.441 |      |
|                                                                                  | taud        | 2.158  | 2.160 | 1.986 | 1.891 | 1.949 | 1.916 | 1.919 | 2.012 | 2.064 | 1.914 | 2.029 | 2.130 |      |
|                                                                                  | Ebn at Noon | 849    | 867   | 819   | 779   | 775   | 741   | 744   | 774   | 789   | 744   | 774   | 817   |      |
|                                                                                  | Edh at Noon | 142    | 149   | 184   | 204   | 190   | 194   | 194   | 178   | 168   | 189   | 161   | 143   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 5.01   | 5.87  | 6.62  | 7.05  | 6.98  | 4.71  | 3.51  | 3.61  | 4.47  | 5.10  | 4.92  | 4.76  |      |
|                                                                                  | RadStd      | 0.15   | 0.14  | 0.22  | 0.18  | 0.34  | 0.48  | 0.42  | 0.26  | 0.39  | 0.27  | 0.27  | 0.13  |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only           | +0.29 | N/A     | N/A    | +0.52   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (3 neighbors) | +0.28 | +0.52   | N/A    | +0.34   | +0.24 | +0.26 | N/A         | N/A     | +160    | +161    |

Nomenclature: See separate page